

LemonVarities

A Comprehensive Image Dataset
for Fresh and Rotten Lemon
Automation

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Dataset Overview

Total Images

~2,000 original lemon images captured with high-resolution smartphone camera

Global Production

9.27 million metric tons of fresh lemons and limes produced worldwide in 2023/2024

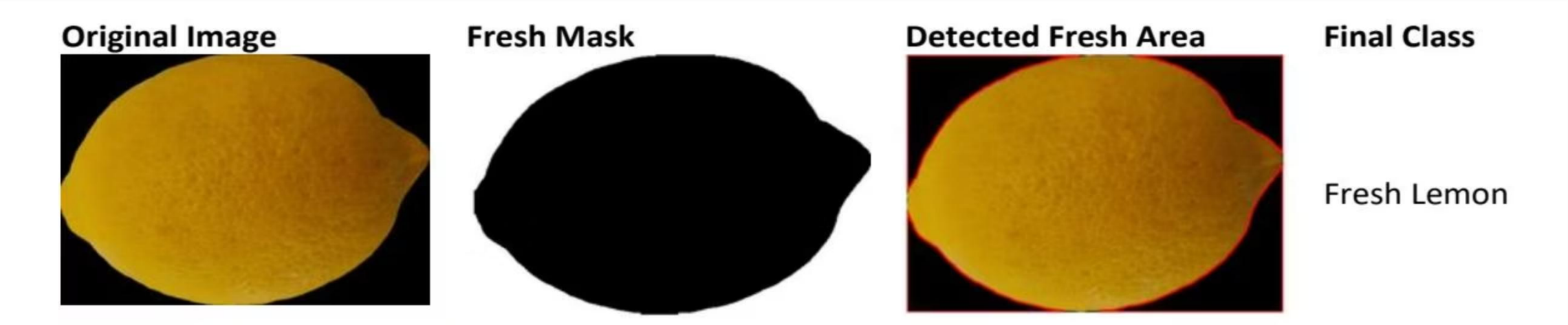
Eight Classes

7 fresh lemon varieties plus 1 rotten lemon class for comprehensive classification

Fresh Lemon Varieties

Seven distinct lemon types collected from Bangladesh and Qatar, each with unique properties and characteristics:

Lemon Type	Length	Weight
Citron	6-8 cm	70-90 g
Eureka	5-6 cm	90-100 g
Meyer	5-6 cm	80-100 g
Desi Kagzi	4-6 cm	60-80 g
Elachi	4-5 cm	70-80 g
Assam	7-9 cm	90-100 g





Fresh Lemon Dataset Details

936

Fresh Images

Original high-quality photographs captured from fields and markets

100%

Field Freshness

Lemons picked directly from fields are completely fresh

95%

Store Freshness

Average freshness of lemons sourced from retail locations

Images captured in various lighting conditions, angles, and backgrounds with standardized 900×1200 pixel dimensions.



Rotten Lemon Classification

1,000 rotten lemon images documenting various decay types for disease detection and quality control:

Fungal Damage

Fungal infections affecting lemon surface and structure

Natural Decay

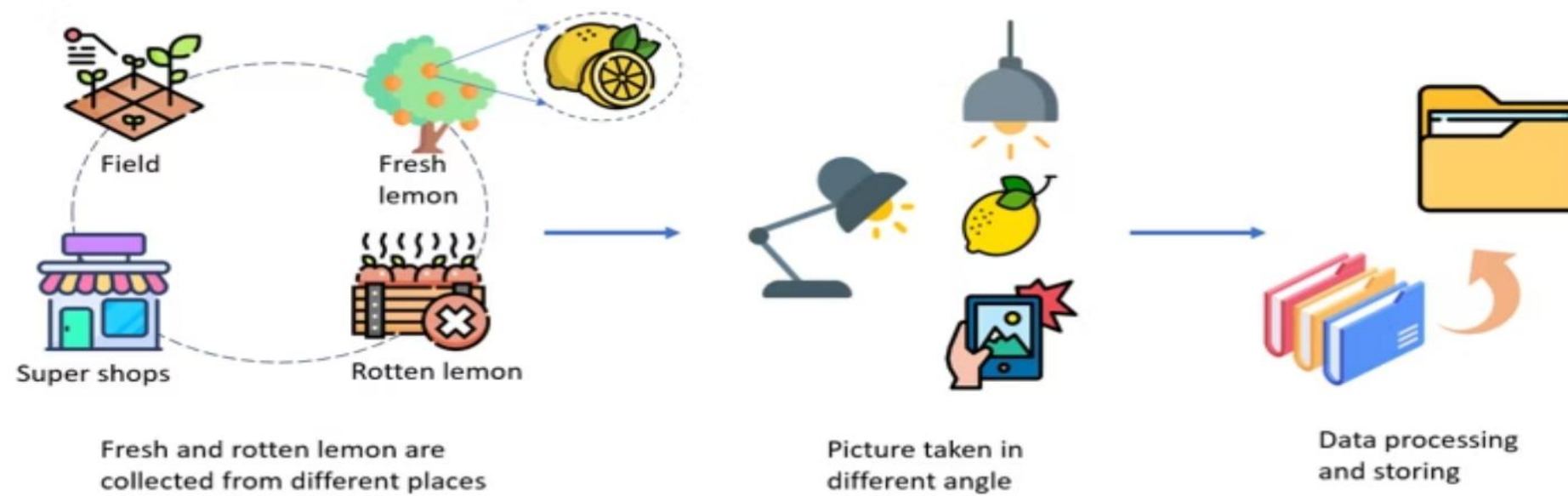
Naturally aged and deteriorated lemons

Microbial Growth

Bacterial and microbial colonization patterns

Physical Damage

Bruising, cuts, and mechanical injury



Data Collection Methodology

01

Source Identification

Traveled to fields, super shops, and rural markets across Bangladesh and Qatar to ensure quality and variety

03

Classification

Properly categorized fresh and rotten lemons, including diseased specimens

02

Expert Verification

Consulted experts to confirm lemon types and authenticity

04

Image Capture

Photographed with Samsung S22 smartphone using optimal lighting and angles

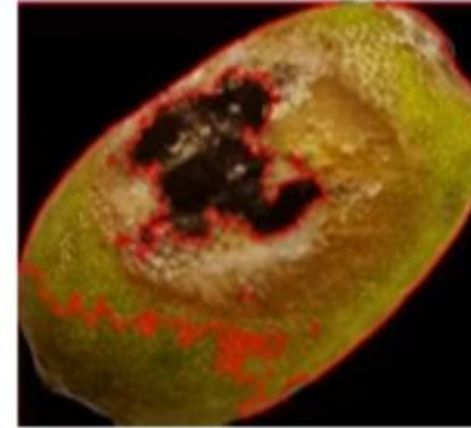
Original Image



Rotten Mask



Detected Rotten area



Final Class

Rotten Lemon

Data Preprocessing & Standardization

Image Processing

- Standardized to 900×1200 pixels
- White background applied consistently
- PNG format for quality preservation
- Brightness compensation applied

Quality Assurance

- Removed blurred and distorted images
- Eliminated distracting backgrounds
- Thorough manual inspection completed
- Final dataset: ~2,000 validated images

Applications & Value



Quality Control

Automate lemon classification in food production and export industries



ML Model Training

Develop AI models for lemon recognition, detection, and real-time classification



Disease Detection

Identify bacterial and viral infections in lemon farms for targeted interventions



Marketing & Branding

High-quality images for product packaging, brochures, and advertising materials

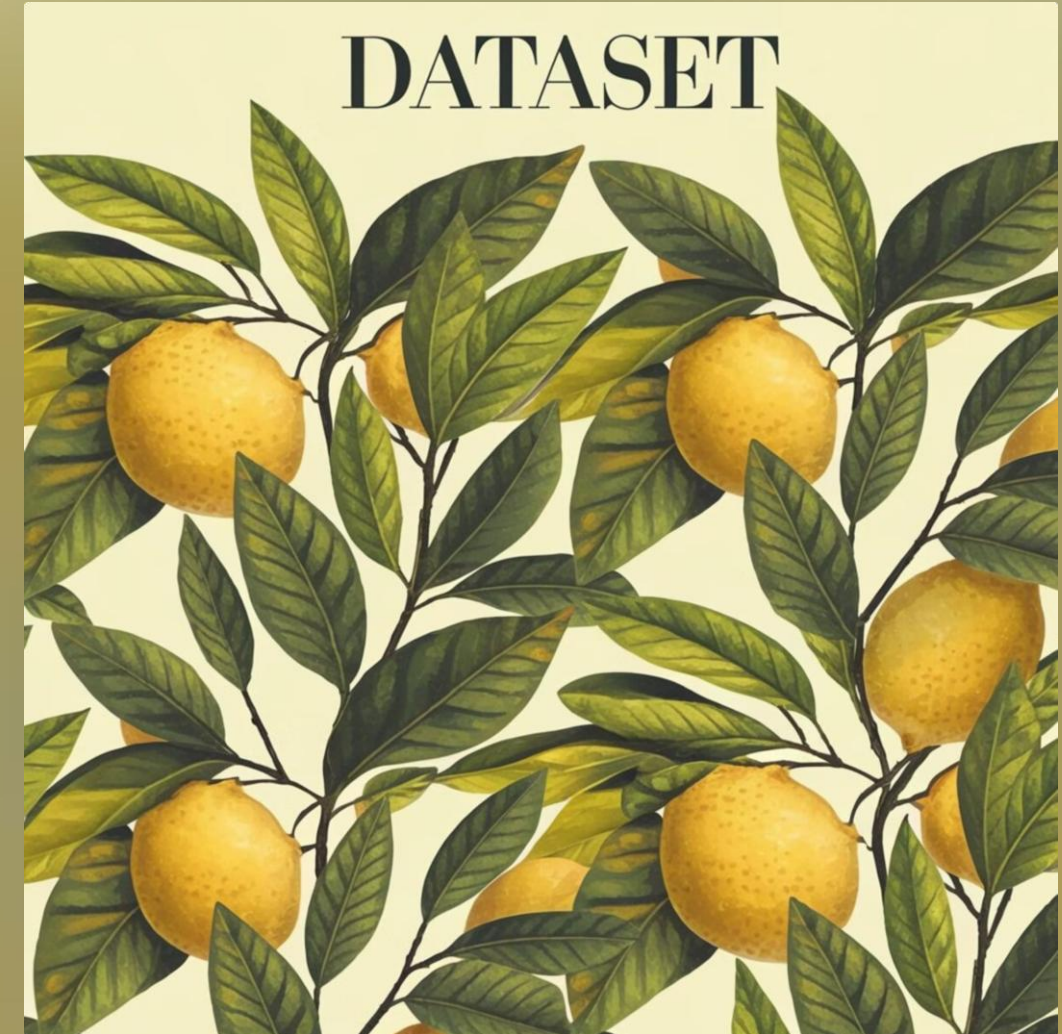
Dataset Specifications & Limitations

Data Specifications

- Fresh lemon: 7 classes
- Rotten lemon: 1 class
- Fresh images: 936
- Rotten images: 1,000
- Resolution: 900×1200 pixels
- Original capture: 3000×4000 pixels
- Format: PNG

Key Limitations

- Only 7 lemon varieties represented
- Limited dataset size due to manual collection
- May not represent all global lemon types
- Specific to Bangladesh and Qatar sourcing



Research Impact & Future Directions

The LemonVarities dataset enables agricultural innovation through machine learning, reducing human effort and production costs while improving quality control across the lemon industry.

- For Farmers
 - Early disease detection and crop management optimization
- For Industry
 - Automated quality assurance and export-ready classification systems
- For Researchers
 - Foundation for advanced AI models in agricultural computer vision

